

University of California, Riverside
Electrical and Computer Engineering Department

Electromagnetics I

Lecture 6: Electromagnetic Waves

Chapter 2

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Office hours: Thursday: 12:00 PM - 1:00 PM, Room: WCH 349

Outline

- James Clerk Maxwell: the discovery
- The meaning of the displacement current
- Electromagnetic waves
- How fast is the speed of light?
- Hertz experiment
- Radio
- Wavefront
- Time-domain (instantaneous) expression

Chapters 2.11 – 2.14

James Clerk Maxwell

1831 – 1879

In 1865, Maxwell published a paper entitled:

A Dynamical Theory of the Electromagnetic Field,

Philosophical Transactions of the Royal Society of London

155, 459-512 (1865).

This is one of the greatest scientific papers ever written.



Maxwell's Equations

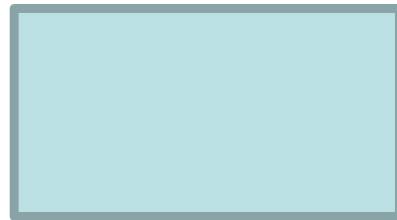
INTEGRAL FORMS

$$\oint_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s} = \int_v \rho_v dv$$

$$\oint_s \mathbf{B} \cdot d\mathbf{s} = 0$$

$$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_s \mathbf{B} \cdot d\mathbf{s}$$

$$\oint_c \mathbf{H} \cdot d\boldsymbol{\ell} = \int_s \mathbf{J} \cdot d\mathbf{s}$$



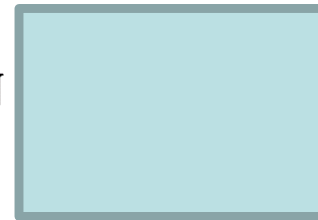
DIFFERENTIAL FORMS

$$\nabla \cdot \epsilon_o \epsilon_r \mathbf{E} = \rho_v \quad (4.1)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (4.2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (4.3)$$

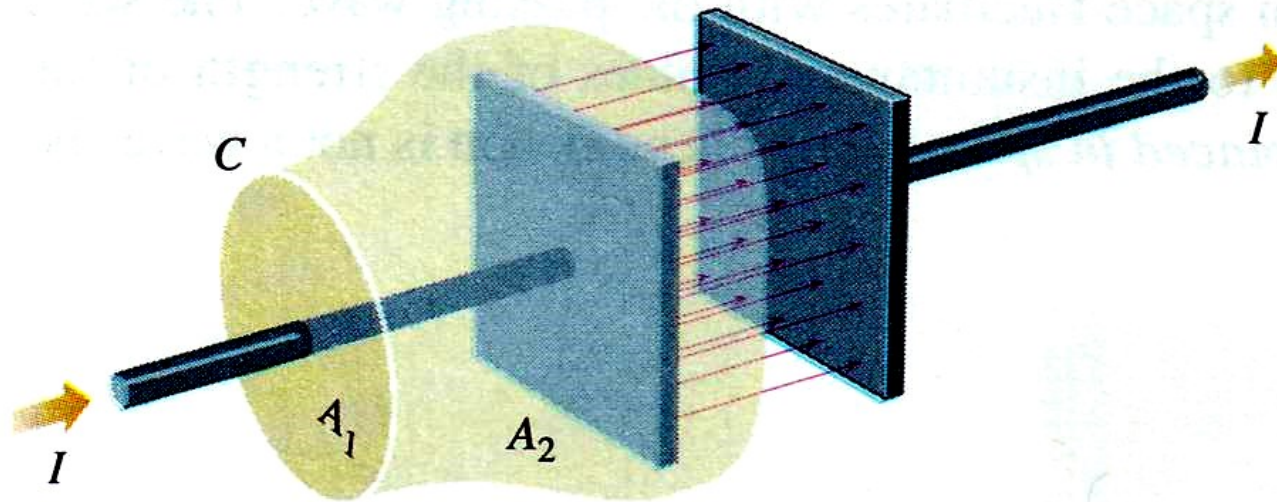
$$\nabla \times \mathbf{H} = \mathbf{J} \quad (4.4)$$



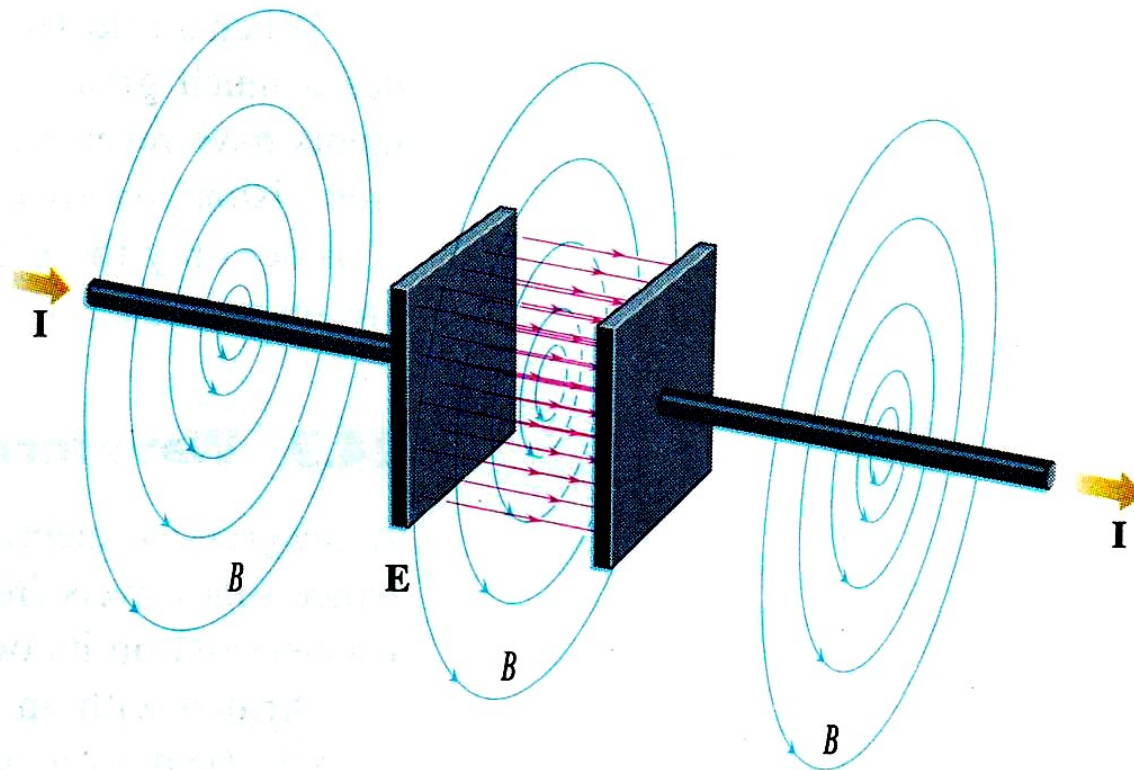
Before Maxwell



Imagine a wire connected to a charging or discharging capacitor. The area in the Amperian loop could be stretched into the open region of the capacitor. In this case there would be current passing through the loop, but not through the area bounded by the loop.



If Ampere's Law still holds, there must be a magnetic field generated by the changing E-field between the plates. This induced B-field makes it look like there is a current (call it the **displacement current**) passing through the plates.



Maxwell's Equations

INTEGRAL FORMS

$$\oint_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s} = \int_v \rho_v dv$$

$$\oint_s \mathbf{B} \cdot d\mathbf{s} = 0$$

$$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_s \mathbf{B} \cdot d\mathbf{s}$$

$$\oint_c \mathbf{H} \cdot d\boldsymbol{\ell} = \int_s \mathbf{J} \cdot d\mathbf{s} + \frac{d}{dt} \int_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s}$$

DIFFERENTIAL FORMS

$$\nabla \cdot \epsilon_o \epsilon_r \mathbf{E} = \rho_v \quad (4.1)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (4.2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (4.3)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \epsilon_o \epsilon_r \mathbf{E}}{\partial t} \quad (4.4)$$

Maxwell introduced



Waves in free space

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\frac{\partial}{\partial t} \vec{\nabla} \times \vec{B} \quad (\text{3rd eq.})$$

$$\vec{\nabla} \times \vec{B} \text{ from 4th eq: } \vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\Rightarrow \vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\text{Vector Identity: } \vec{\nabla} \times \vec{\nabla} \times \vec{E} = \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E}$$

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \quad \text{-Laplacian operator}$$

$$\nabla^2 E_x = \frac{\partial^2 E_x}{\partial x^2} + \frac{\partial^2 E_x}{\partial y^2} + \frac{\partial^2 E_x}{\partial z^2}$$

$$\text{from 1st eq: } \vec{\nabla} \cdot \vec{E} = 0 \Rightarrow \vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\nabla^2 \vec{E}$$

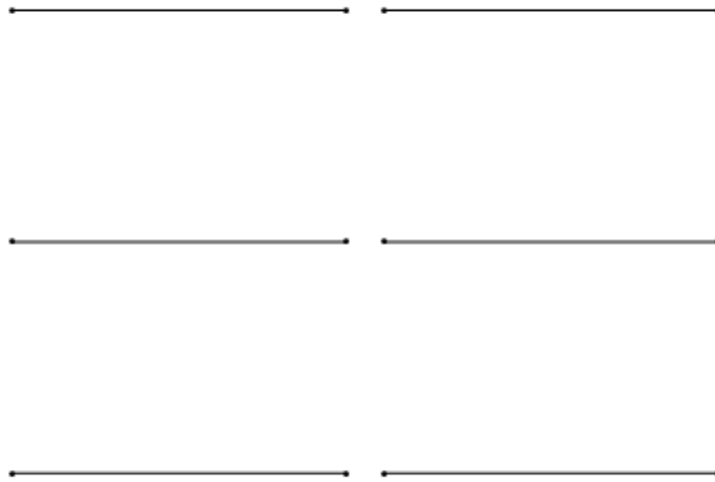
$$-\nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{E} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

- homogeneous vector wave equation

Wave Equation



Transverse waves on a string obey the equation $\frac{\partial^2 u}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 u}{\partial t^2}$ where $u(x,t)$ is the transverse displacement at position x , v is the wave speed, x is the position and t the time.

Manipulation of Maxwell's equations leads to the following plane **wave equations** for $\vec{E}$ and $\vec{B}$:

Equations on this slide are for waves propagating along x-direction.

$$\frac{\partial^2 E_y}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_y(x, t)}{\partial t^2}$$

$$\frac{\partial^2 B_z}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 B_z(x, t)}{\partial t^2}$$

These equations have solutions:

$$E_y = E_{\max} \sin(kx - \omega t)$$

$$B_z = B_{\max} \sin(kx - \omega t)$$

$E_{\max}$ and $B_{\max}$ are the electric and magnetic field amplitudes

where

$$k = \frac{2\pi}{\lambda},$$

$$\omega = 2\pi f,$$

and

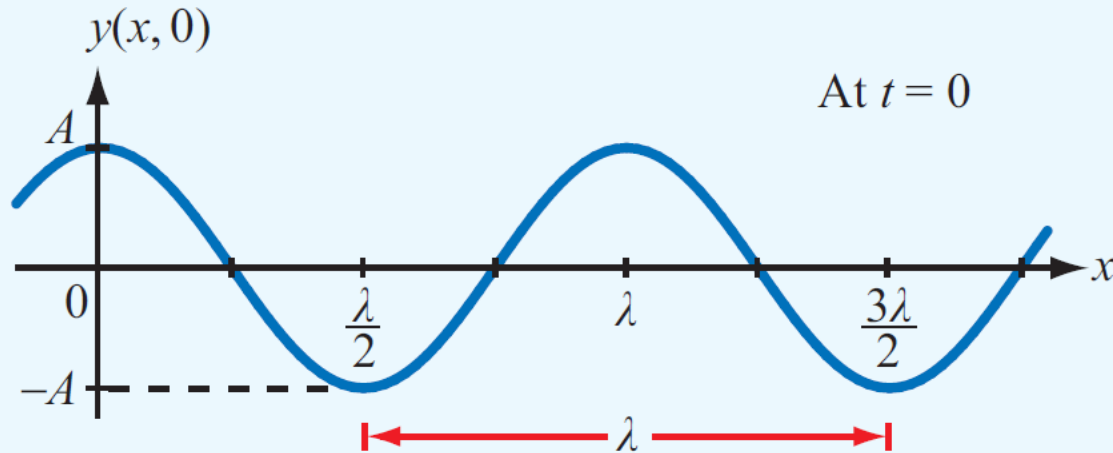
$$f\lambda = \frac{\omega}{k} = c.$$

You can verify this by direct substitution.

$E_{\max}$ and $B_{\max}$ in these notes are sometimes written by others as E_0 and B_0 .

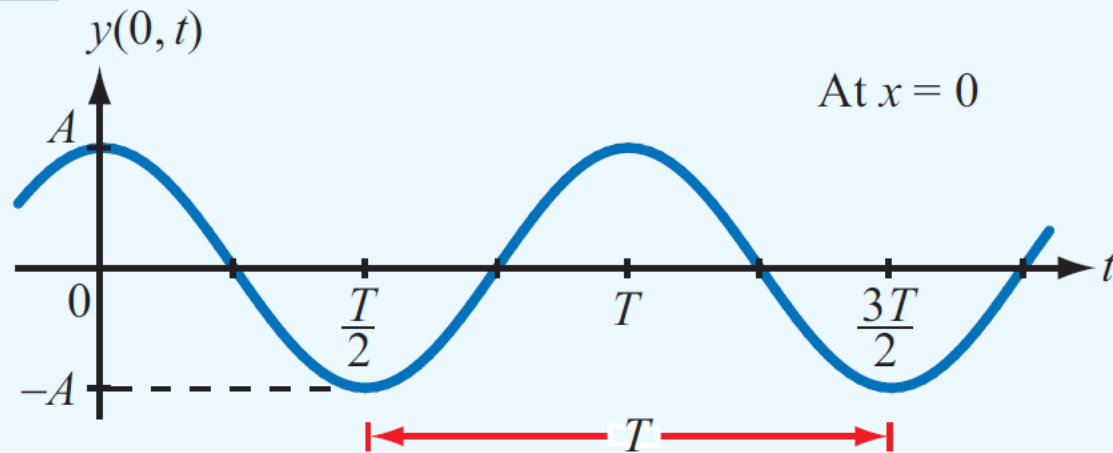
Sinusoidal wave

$$y(x, t) = A \cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0\right)$$



(a) $y(x, t)$ versus x at $t = 0$

$$u_p = \frac{dx}{dt} = \frac{\lambda}{T} \quad (\text{m/s})$$



(b) $y(x, t)$ versus t at $x = 0$

$$u_p = f\lambda \quad (\text{m/s})$$

Maxwell's Discovery The Speed of Light

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

*The speed of light was determined by many scientists using many different methods
but this determination by Maxwell was serendipitous!*

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = \frac{1}{\sqrt{(8.85 \times 10^{-12} \frac{C}{Nm^2})(4\pi \times 10^{-7} \frac{N}{A^2})}} = 3.00 \times 10^8 \text{ m/s}$$

experimentally determined constants

How fast is the speed of light?



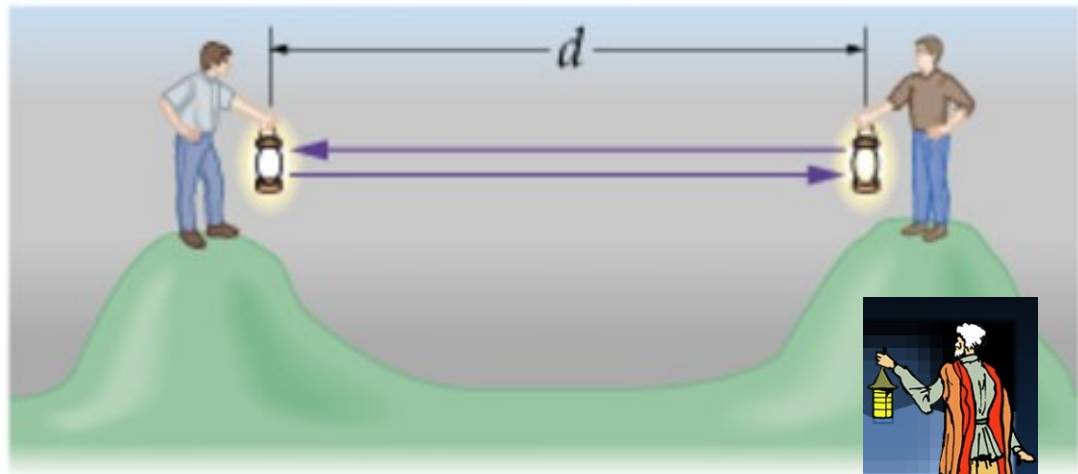
1667 Lantern Experiment



- The Accademia del Cimento of Florence took Galileo's suggestion and made the first attempt to actually measure the velocity of light.
 - Two people, A and B, with covered lanterns went to the tops of hills about 1 mile apart.
 - First A uncovers his lantern. As soon as B sees A's light, he uncovers his own lantern.
 - Measure the time from when A uncovers his lantern until A sees B's light, then divide this time by twice the distance between the hill tops.
- Therefore, the speed of light would theoretically be $c = (2D)/t$.

■ Human reaction times are approx. 0.2 sec and therefore, too slow to determine c with any accuracy.

■ Proved speed of light was finite and showed that light travels at least 10x faster than sound.

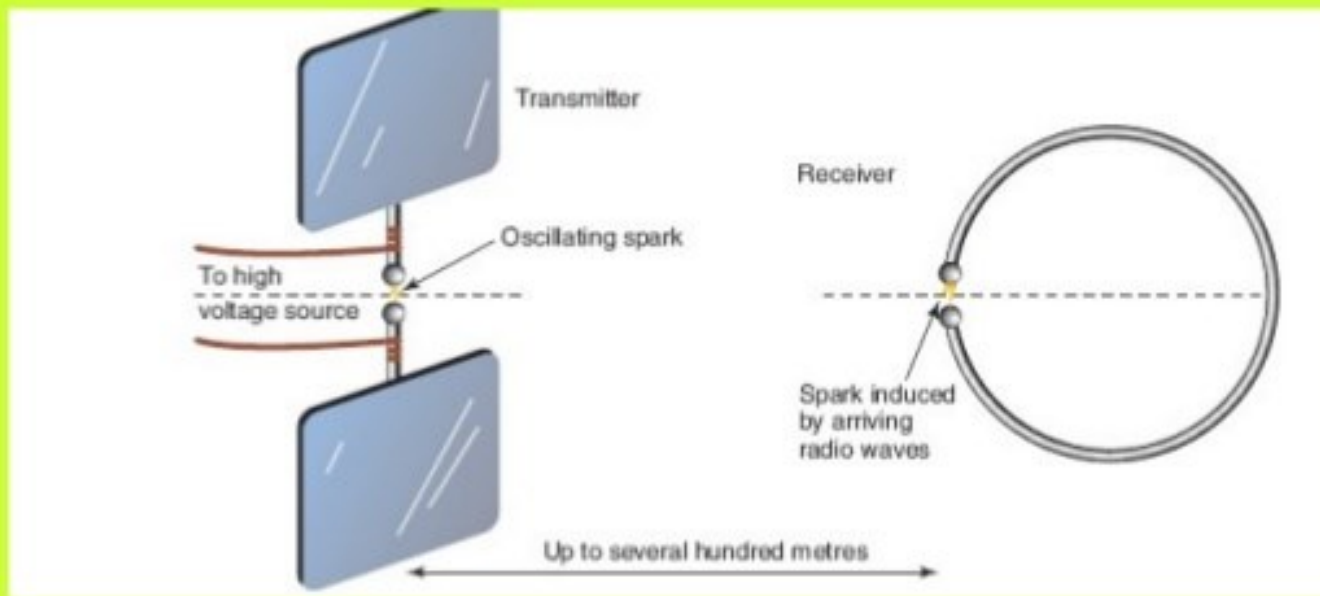


Historical Accuracy of speed of light

Date	Experimentor	Country	Experimental Method	Speed (10 ⁸ m/s)	Uncertainty (m/ true c	Error from
1600	Galileo	Italy	Lanterns and shutters	"Fast"	?	
1676	Roemer	France	Moons of Jupiter	2.14	?	28%
1729	Bradley	England	Aberration of Light	3.08	?	2.70%
1849	Fizeau	France	Cog Wheel	3.14	?	4.70%
1879	Michelson	United States	Rotating mirror	2.9991	75000.0	400 in 10 ⁶
	Michelson	United States	Rotating mirror	2.99798	22000.0	18 in 10 ⁶
1950	Essen	England	Microwave cavity	2.997925	1000.0	0.1 in 10 ⁶
1958	Froome	England	Interferometer	2.997925	100.0	0.1 in 10 ⁶
1972	Evenson et al.	United States	Laser Method	2.99792457	1.1	2 in 10 ⁹
1974	Blaney et. al	England	Laser Method	2.99792459	0.6	3 in 10 ⁹
1976	Woods et al.	England	Laser Method	2.99792459	0.2	3 in 10 ⁹
1983		International		2.99792458	0.0	Exact

The Hertz Experiments: 1886-1888

- Hertz used a rapidly oscillating spark (charge) in an induction coil to produce a rapidly changing electric field (creating a changing magnetic field...). This sent EM waves across a distance of a few hundred metres, where they were detected by a receiving loop.



The breakdown voltage of air is about 3kV/mm

Hertz [Hz] is a unit of frequency

Top to bottom:

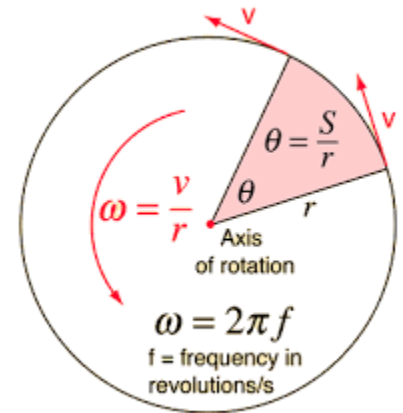
● $f = 0.5 \text{ Hz}$
 $T = 2.0 \text{ s}$

Lights flashing at frequencies $f = 0.5 \text{ Hz}$, 1.0 Hz and 2.0 Hz , i.e. at 0.5, 1.0 and 2.0 flashes per second, respectively.

● $f = 1.0 \text{ Hz}$
 $T = 1.0 \text{ s}$

The time between each flash – the *period* T – is given by $1/f$ (the reciprocal of f), i.e. 2, 1 and 0.5 seconds, respectively.

● $f = 2.0 \text{ Hz}$
 $T = 0.5 \text{ s}$



Transistors can be switched On and Off at the GHz frequencies

1 GHz = 1 000 000 000 Hz

Novelty Stage

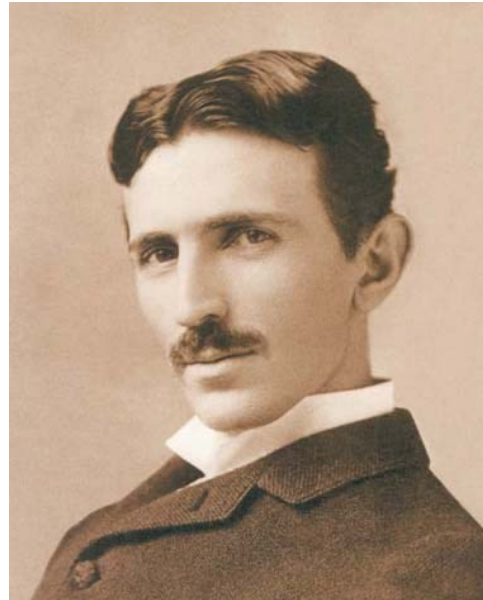
THREE GUYS WHO INVENTED THE DEVICE TO TRANSMIT A SIGNAL THROUGH THE AIR (1894)

A. Guglielmo Marconi vs.

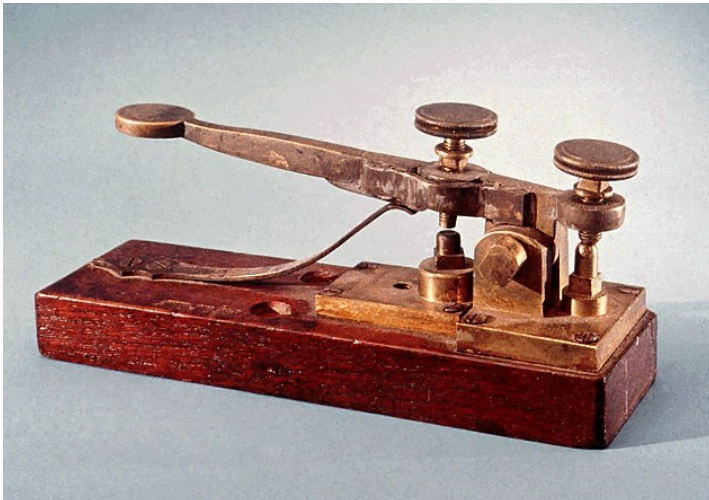
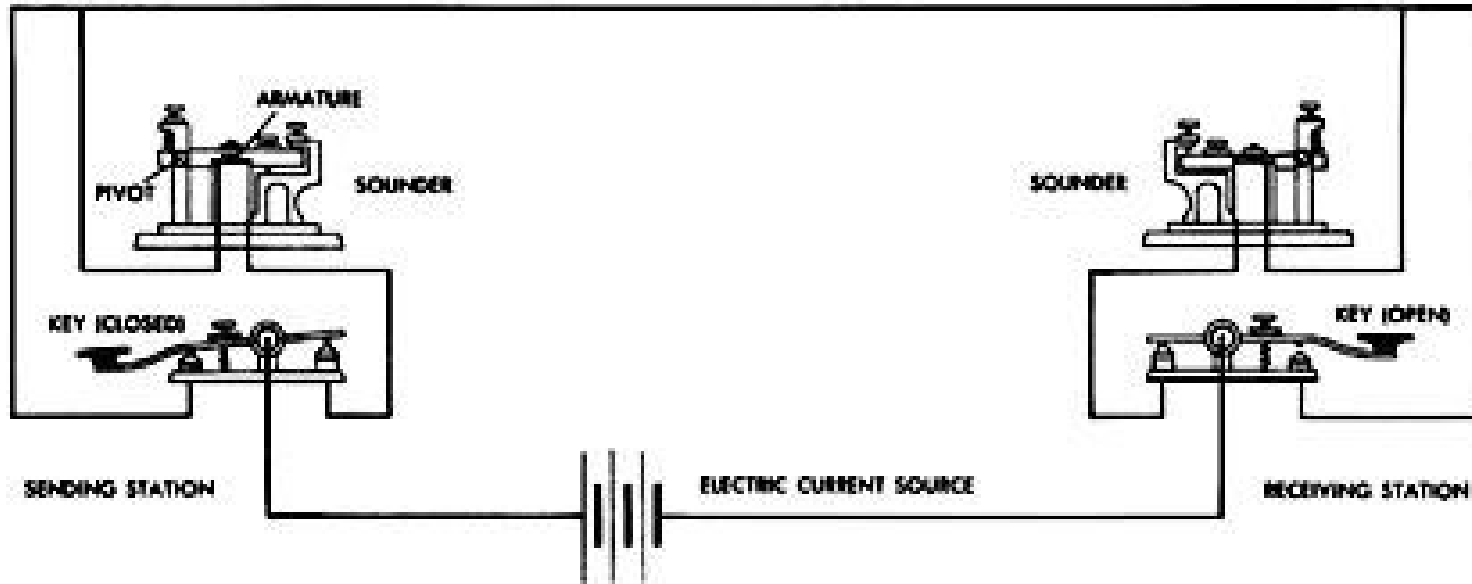
B. NiKola Tesla vs.

C. Alexander Popov

– Who invented Wireless telegraphy?



It is started with Telegraph

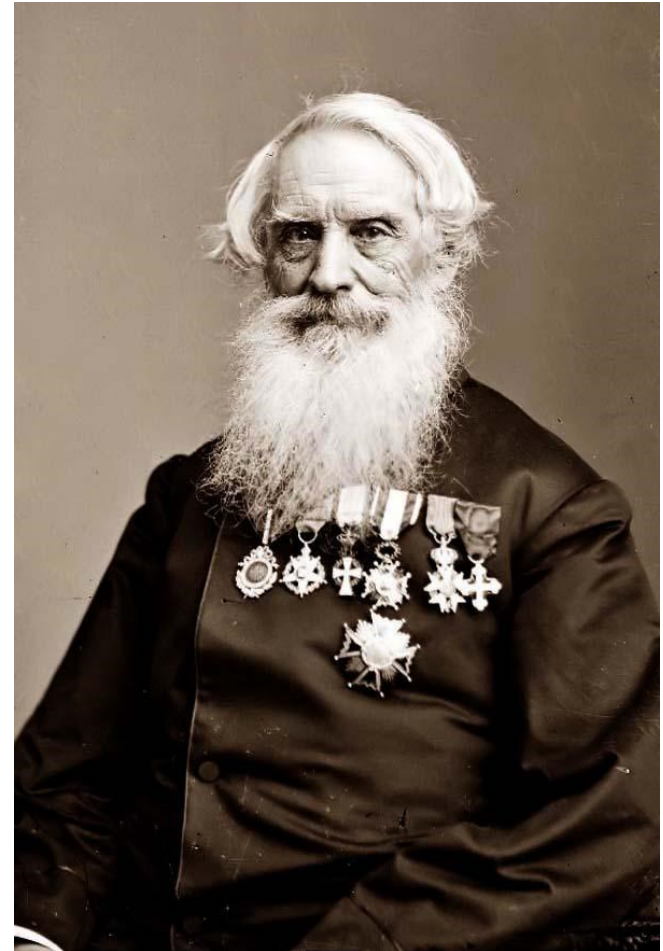


- Used a an on-off switch – the telegraph key – to turn an electric current on and off, sending pulses of current through a wire to an electromagnet that would click in time to the pulses

Morse Code Alphabet

The International morse code characters:

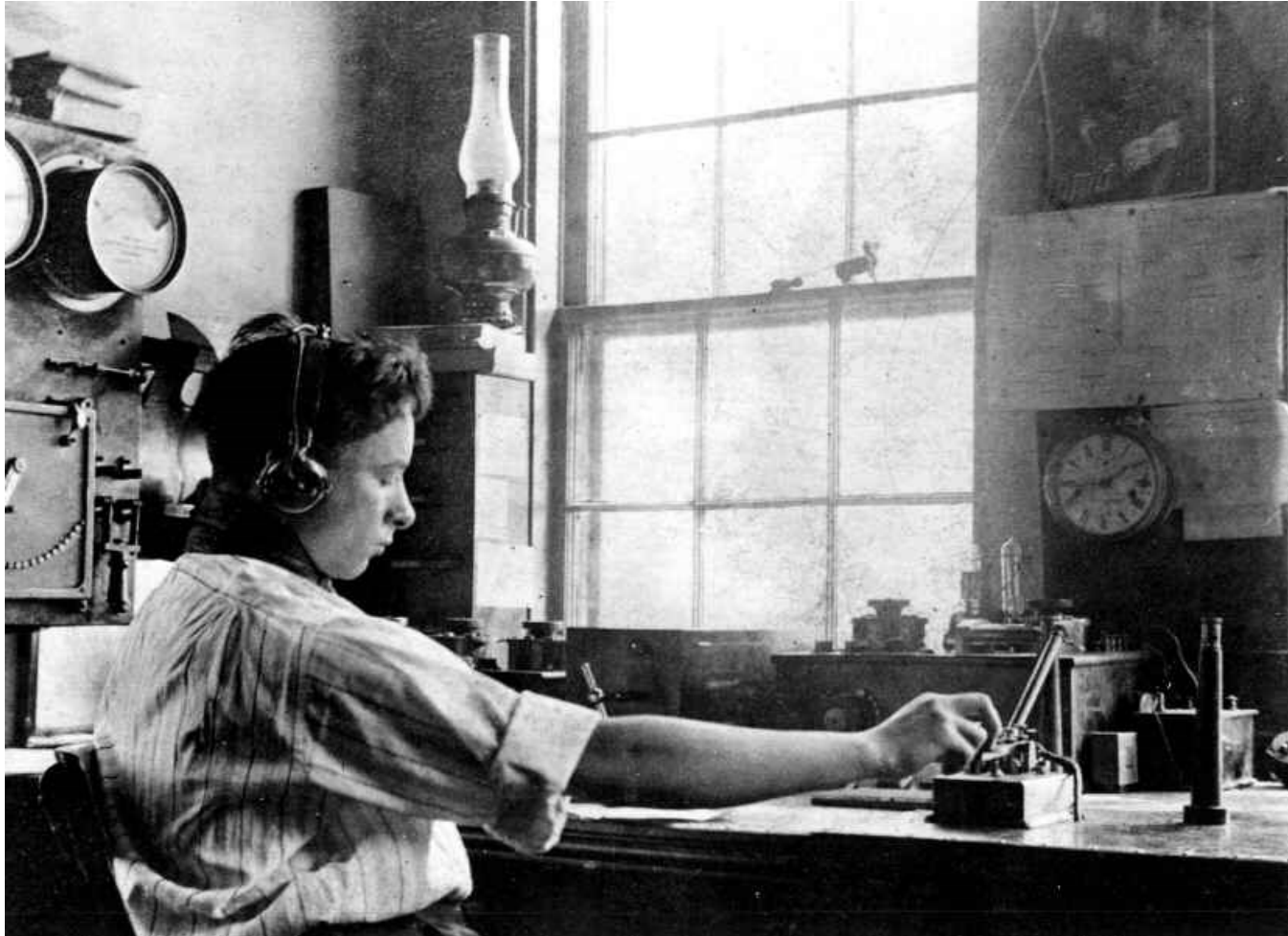
A .-	N -.	0 -----
B -...	O ---	1 .----
C -.-.	P .-..	2 ..---
D -..	Q --.-	3 ...--
E .	R .-.	4-
F ..-.	S ...	5
G --.	T -	6 -....
H	U ..-	7 --...
I ..	V ...-	8 ---..
J .----	W .--	9 ----.
K -.-	X -.-	Fullstop .----
L .-..	Y -.-	Comma --.---
M --	Z --..	Query ..----



Portrait of Samuel F. B. Morse taken by Mathew Brady, in 1866. Medals worn (from wearer's right to left, top row: Nichan Iftikhar (Ottoman); Order of the Tower and Sword (Portugal); Order of the Dannebrog (Denmark); Gold Medal of Art and Science (Württemberg); Legion of Honour (France); Order of Saints Maurice and Lazarus (Italy). Bottom row: Order of Isabella the Catholic (Spain)

Entrepreneurial Stage

- David Sarnoff and the Titanic

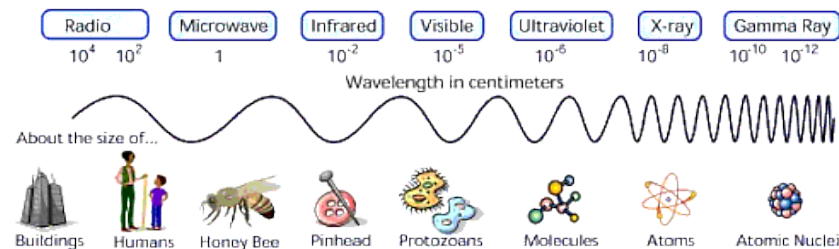
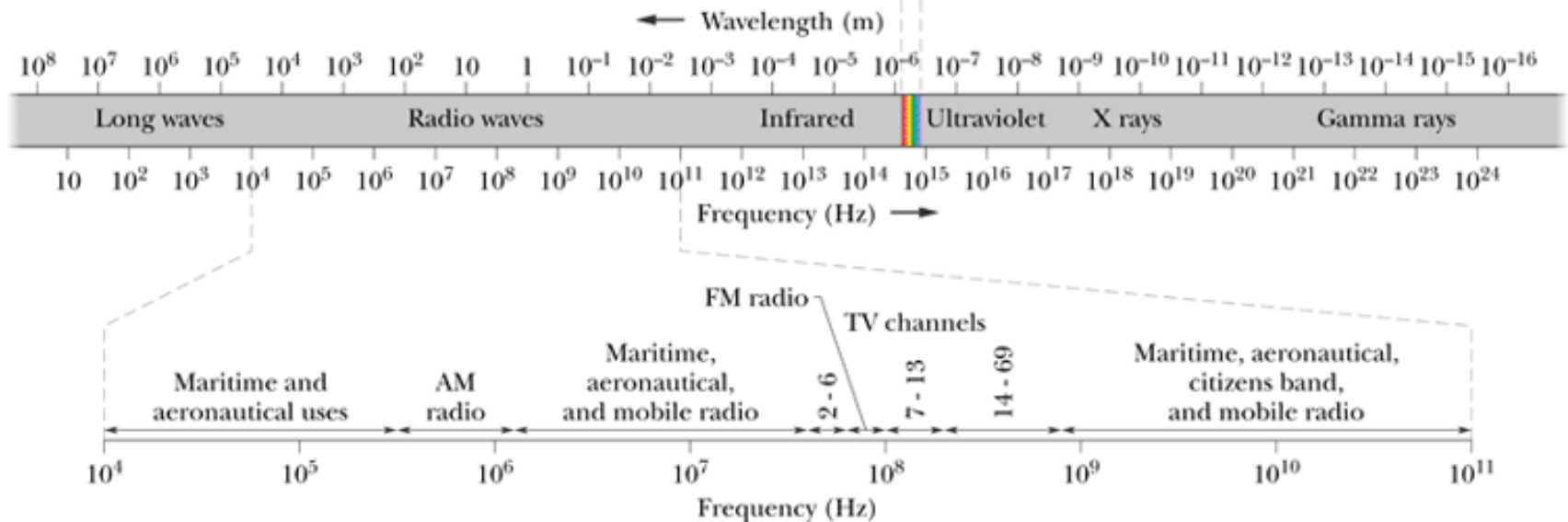
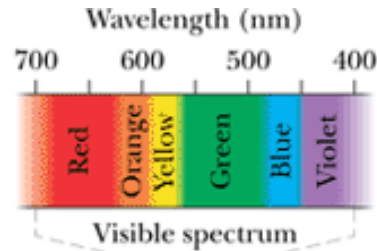


Still using Morse code with radio

April 15, 1912

Maxwell's Rainbow

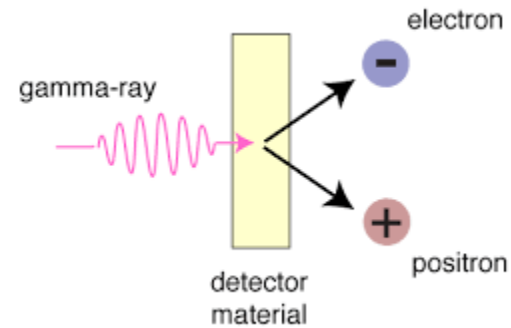
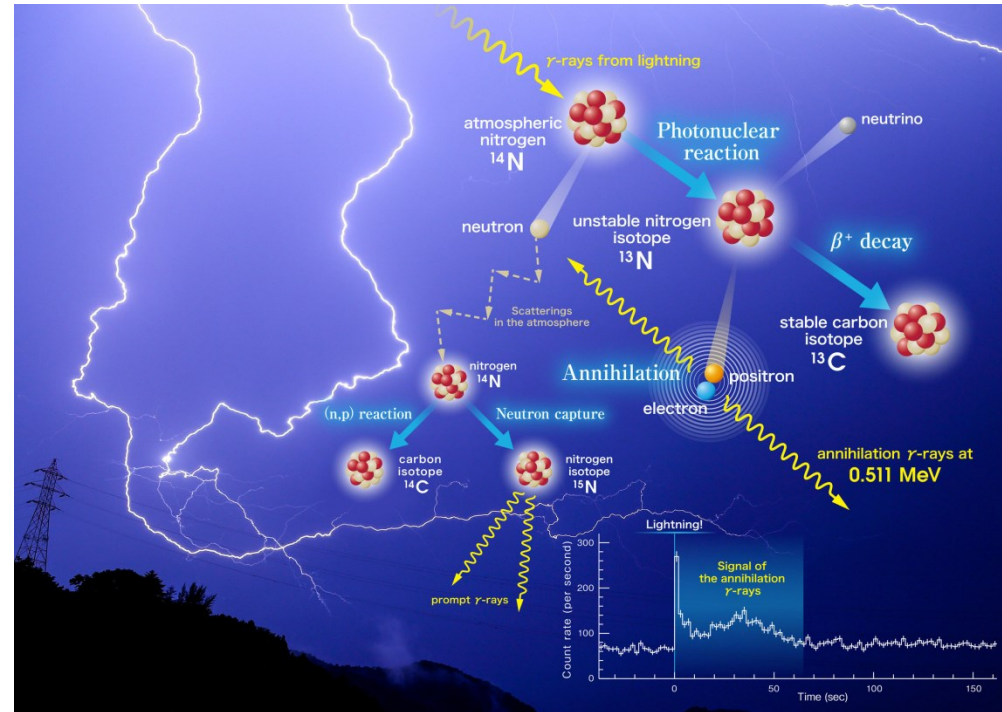
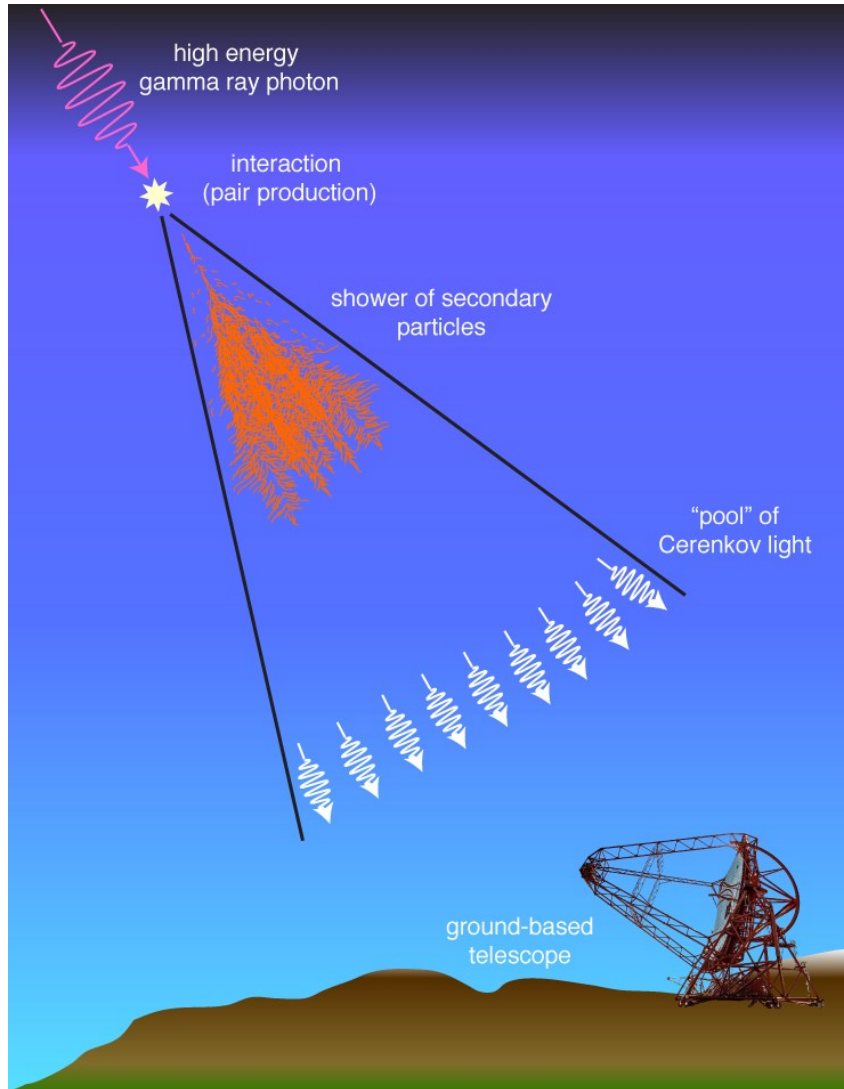
Extremely low frequency (ELF) is the ITU designation for electromagnetic radiation (radio waves) with frequencies from 3 to 30 Hz, and corresponding wavelengths of 100,000 to 10,000 kilometers, respectively). ELF waves can penetrate seawater, which makes them useful in communication with submarines. The US, Russia, India, and China are the only nations known to have constructed ELF communication facilities.



VLF Communication

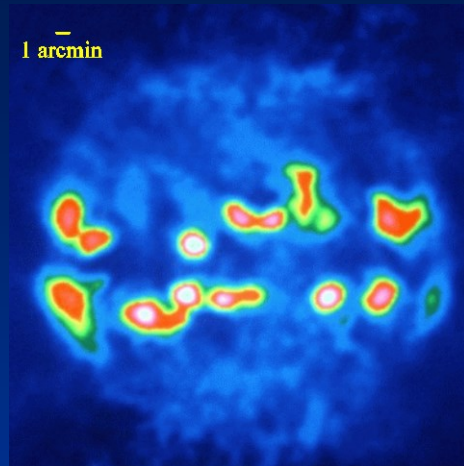


Detecting Gamma Rays

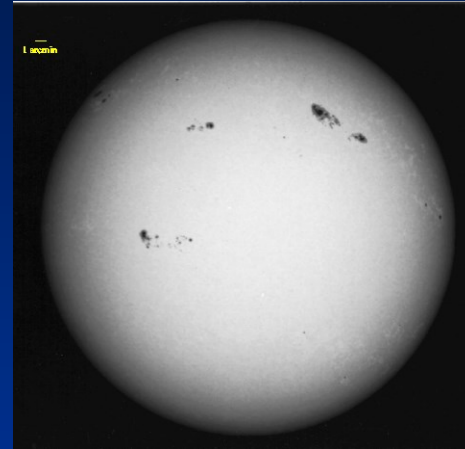


The Sun in different “light”

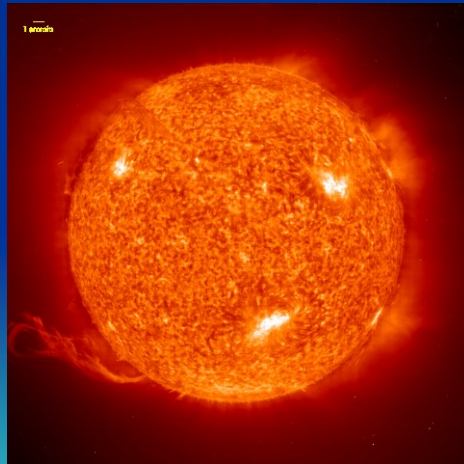
Radio



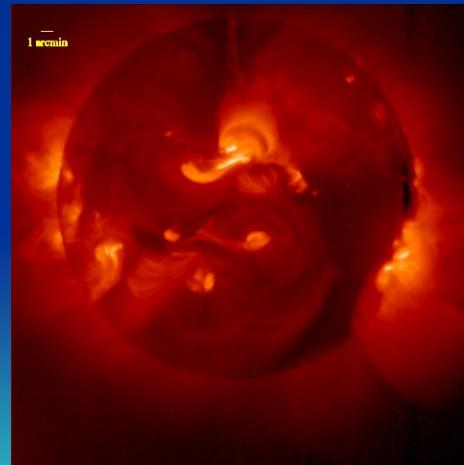
Visible



Ultraviolet

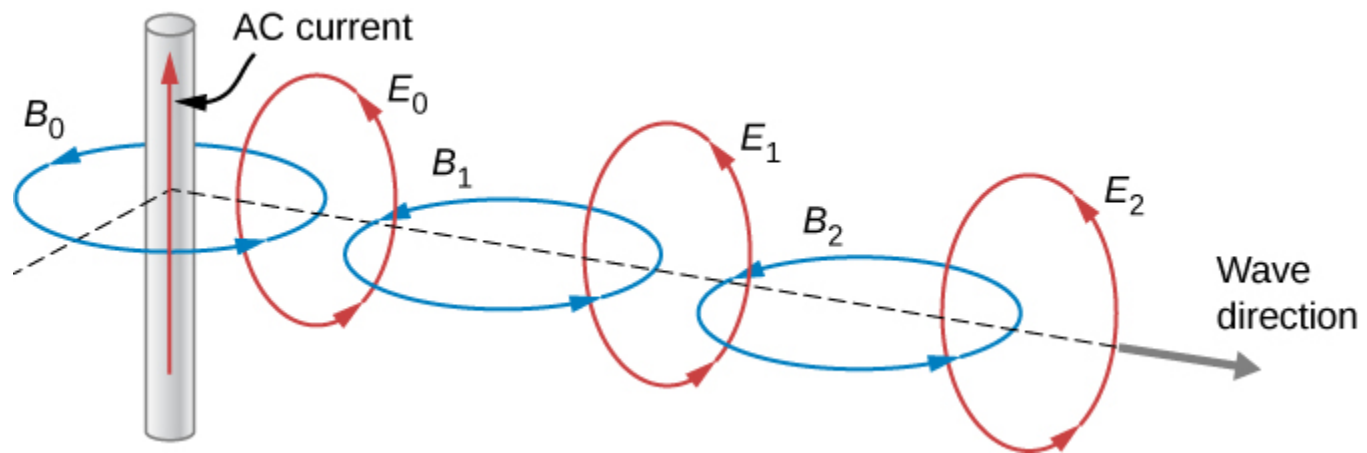
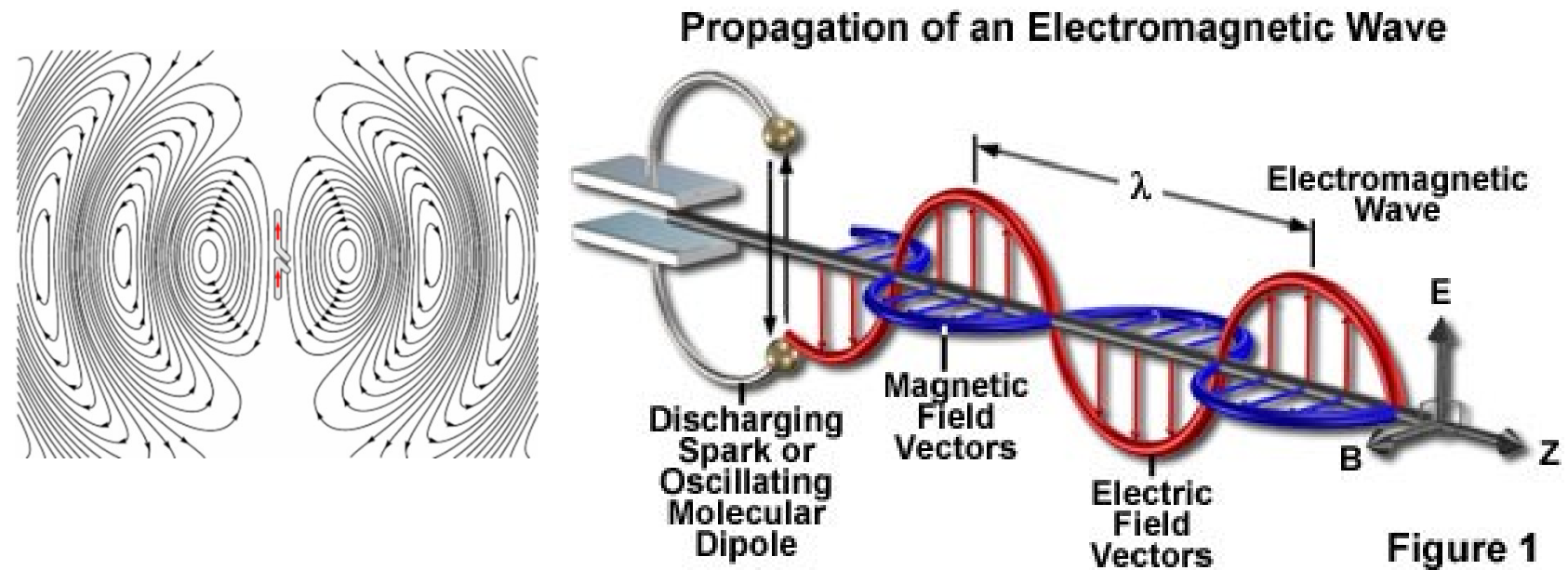


X-Ray

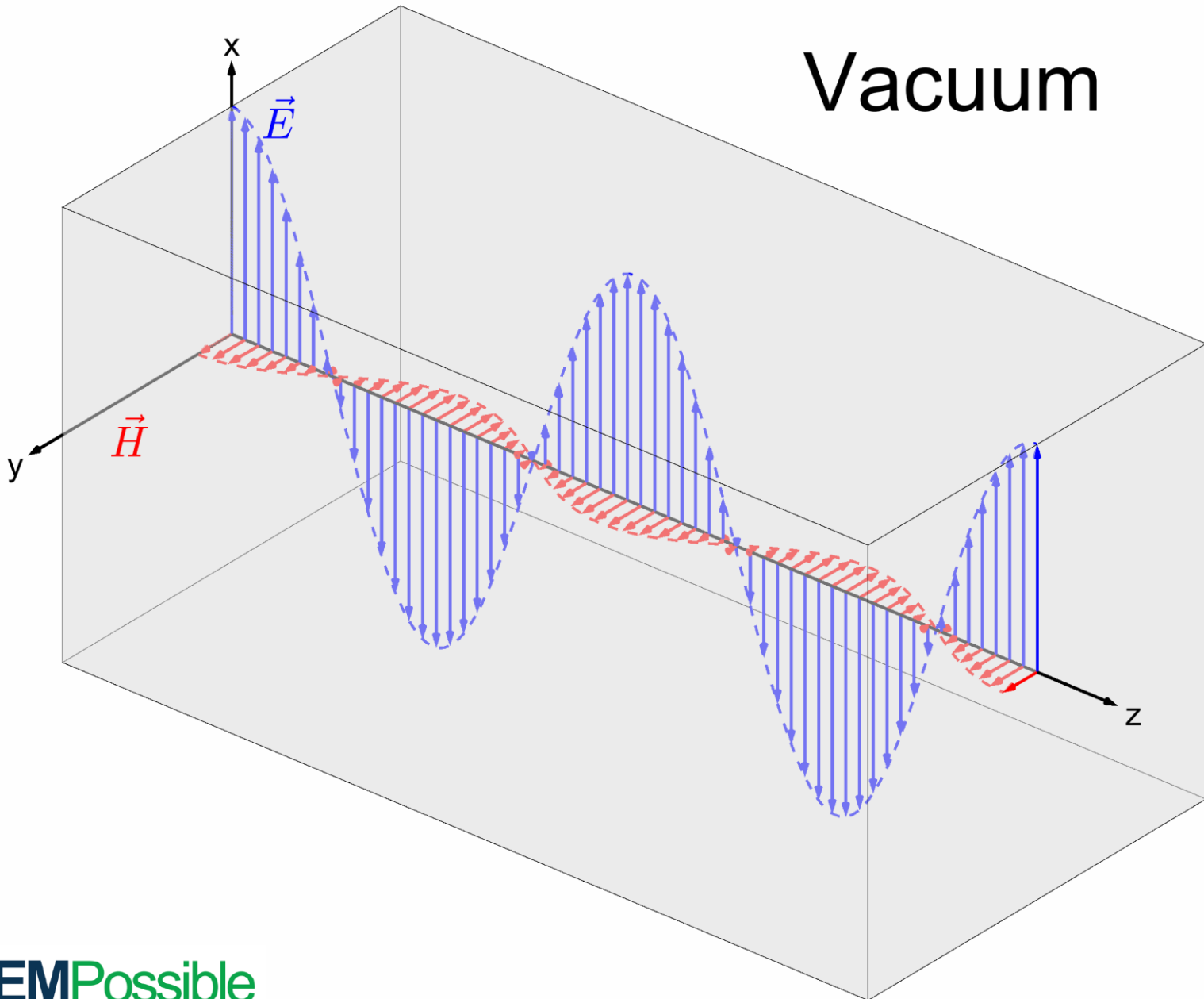


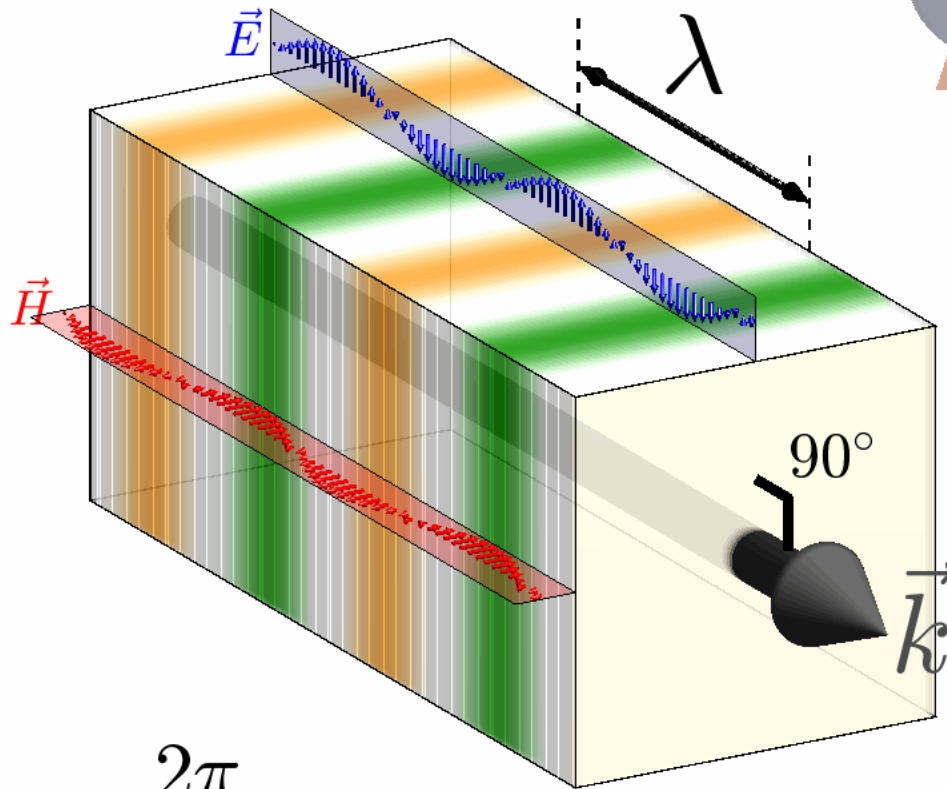
Images courtesy of: NRAO/AUI/NSF/G. Dulk, D. Gary (radio), NSO/AURA/NSF (visible) SOHO/NASA (ultraviolet) and Yohkoh/ISIS/NASA (X-Ray)

Propagating Electromagnetic Wave



Vacuum





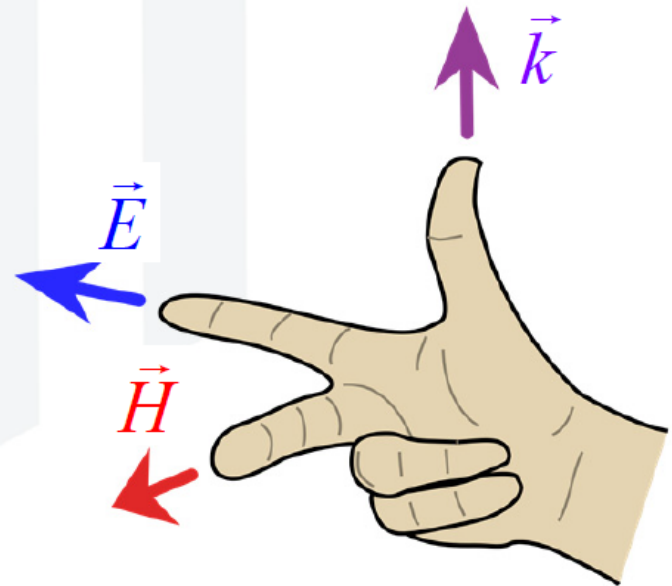
$$|\vec{k}| = \frac{2\pi}{\lambda}$$

Plane Waves in LHI Media are TEM

In linear, homogeneous, and isotropic (LHI) media, the electric field $\vec{E}$, magnetic field $\vec{H}$, and direction of the wave $\vec{k}$ are all perpendicular to each other. These are called transverse electromagnetic (TEM) waves.

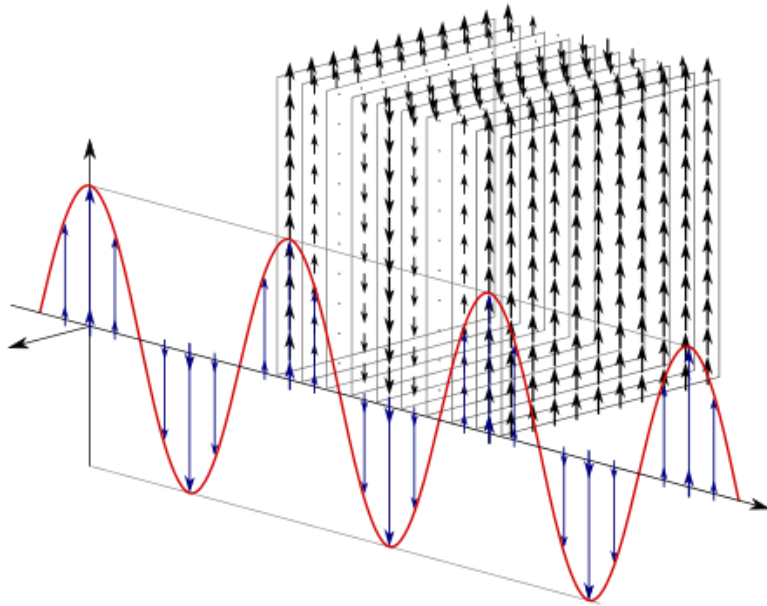
$$\vec{E} \perp \vec{k} \perp \vec{H}$$

Further $\vec{E}$, $\vec{H}$, and $\vec{k}$ follow a right-hand rule.

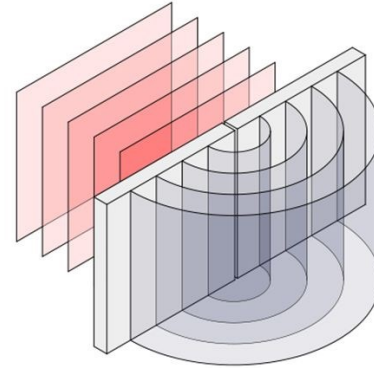


Plane, Cylindrical, and Spherical Electromagnetic Waves

Plane wave



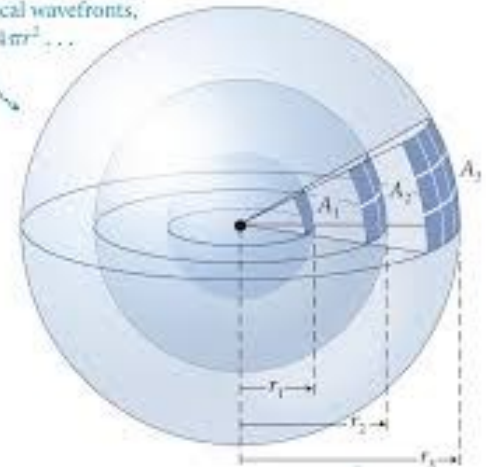
Cylindrical waves



$$\Psi = \left(\frac{A}{\sqrt{\rho}} \right) e^{i(k\rho \pm \omega t)}$$

Spherical Wave

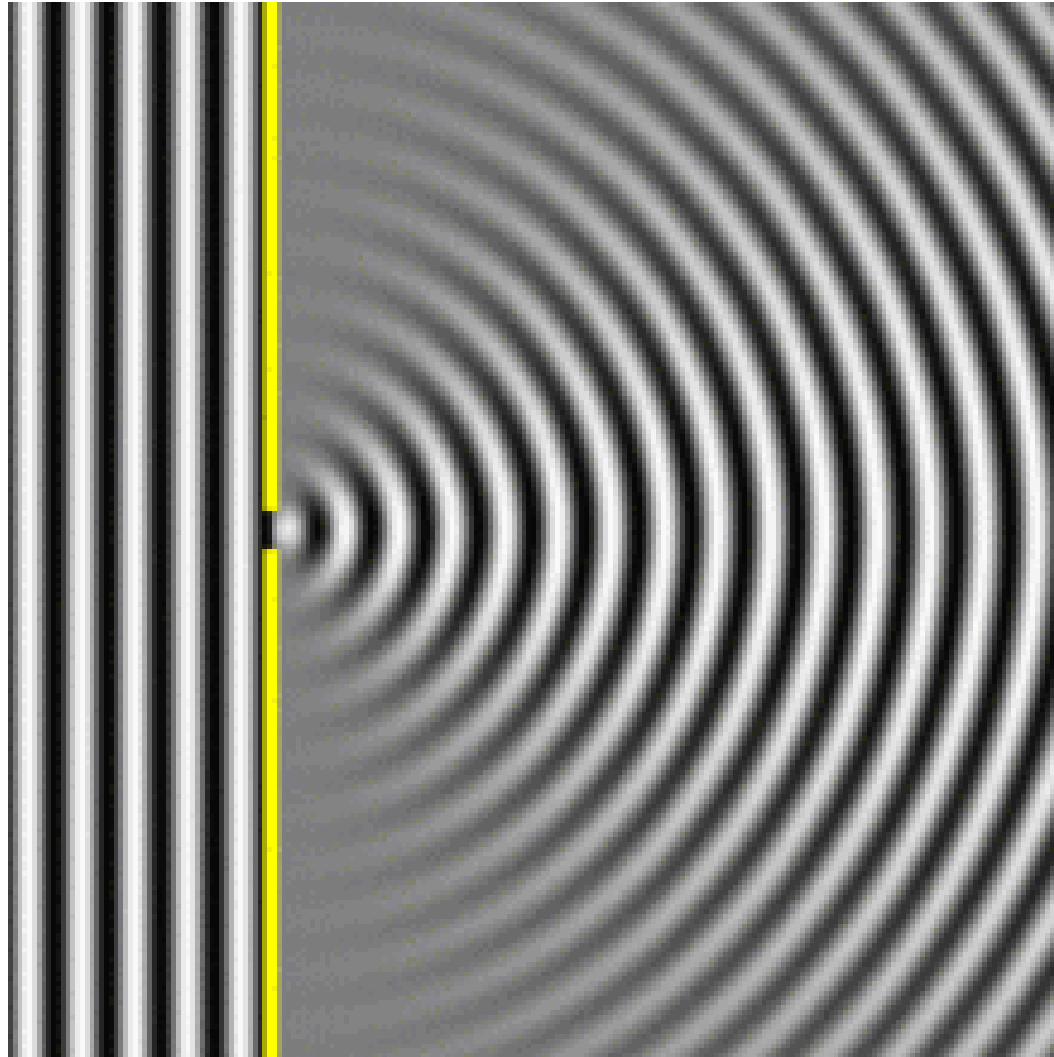
For spherical wavefronts,
area $A = 4\pi r^2 \dots$



$\dots$ so energy per unit area decreases as $1/r^2$
(inverse-square relationship).

In physics, a wavefront is the locus of points characterized by propagation of positions of identical phase: propagation of a point in 1D, a curve in 2D or a surface in 3D. For an electromagnetic wave, the wavefront is represented as a surface of identical phase, and can be modified with conventional optics. For instance, a lens can change the shape of optical wavefronts from planar to spherical as the lens introduces a spatial phase variation across the beam shape.

Plain to spherical



Electro-magnetic waves

time-domain (instantaneous) expression

$$\vec{E}(z, t) = E_{max} \cos(\overbrace{2\pi \times f t}^{\omega} - \underbrace{\beta_0 z}_{\frac{2\pi}{\lambda}} + \phi_0) \vec{a}_x \quad \text{V/m}$$

$$\vec{H}(z, t) = H_{max} \cos(2\pi \times f t - \beta_0 z + \phi_0) \vec{a}_y \quad \text{A/m}$$

- By defining

radian frequency = $\omega = 2\pi f$ (rad/s)

wavenumber = $\beta = \frac{2\pi}{\lambda}$ (rad/m)

Note: Radian frequency is also denoted *angular velocity* and wavenumber is also denoted *phase constant*

Reading the time-domain expression

$$f\lambda = \frac{\omega}{k} = c$$

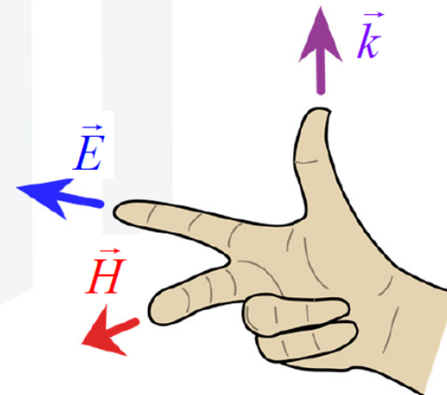
all electromagnetic waves propagate with the same speed in vacuum

Plane Waves in LHI Media are TEM

In linear, homogeneous, and isotropic (LHI) media, the electric field $\vec{E}$, magnetic field $\vec{H}$, and direction of the wave $\vec{k}$ are all perpendicular to each other. These are called transverse electromagnetic (TEM) waves.

$$\vec{E} \perp \vec{k} \perp \vec{H}$$

Further $\vec{E}$, $\vec{H}$, and $\vec{k}$ follow a right-hand rule.



Summary

- Displacement current is a quantity introduced by James Clerk Maxwell, defined as the rate of change of the electric field that induces a magnetic field, acting like current between capacitor plates. It resolves inconsistencies in Ampère's law by ensuring charge conservation, which is essential for describing electromagnetic waves.
- Electromagnetic (EM) waves consist of oscillating electric and magnetic fields.
- EMs may be Plane, Cylindrical, and Spherical
- The time-domain (instantaneous) expression represents a signal's value as it changes over time.
- One can extract the frequency, the phase constant, the direction of propagation, and the amplitudes of the electric and magnetic fields by reading the instantaneous expression.